Dear Editor,

Please consider my manuscript entitled:

"Machine Learning Assisted Analysis of a Static Relativistic Gravitational Potential: A Comparative Study with Newtonian Approximation and Error Quantification"

for publication as a **Full Length Article** in the **Journal of Computational Science**.

In this work, I present a computational framework for analyzing a static relativistic gravitational potential by integrating analytical modeling, numerical error analysis, and machine learning. The study quantitatively compares the relativistic potential with its Newtonian approximation, identifies the limits of the classical approximation through error quantification, and evaluates multiple machine learning algorithms as surrogate models for approximating the underlying physical relationship.

The manuscript combines concepts from computational physics, numerical analysis, and scientific machine learning to demonstrate how data-driven methods can complement analytical physical models while preserving physical interpretability. The work aims to contribute to the growing field of computational science by illustrating the integration of modern machine learning techniques with established theoretical physics.

I confirm that this manuscript represents original work, has not been published previously, and is not under consideration for publication elsewhere. I have read and agree to the policies of the journal.

I believe this work will be of interest to the readers of the **Journal of Computational Science**, particularly researchers working in computational physics, scientific machine learning, numerical modeling, and computational methods for physical sciences.

Thank you for your time and consideration. I look forward to your response.

Sincerely,

Aayush Kumar
Corresponding Author